A multi-omics data brokering approach from DataHub to ELIXIR Deposition Databases

ELIXIR All Hands, 5-8 June 2023, Dublin, Ireland

Brokering data from data producers to data repositories is an integral service for research data management (RDM) platforms. However, existing tools are often technique/domain-specific and focused on a single repository, thus not reflecting the structure of scientific projects making use of multiple technologies (multi-omics) in the same study. Here, we propose a unified and domain/technique-agnostic data brokering approach that can submit metadata and data from multi-omics studies to multiple specialised repositories. Specifically, we use the FAIRDOM-SEEK based platform DataHub to collect metadata according to repositories' checklists and to the ISA framework, an established system for capturing the heterogeneous metadata from multi-omics studies. The proposed approach can reduce the need for extensive manual curation and mapping of (meta)data and enable researchers to publish their data more efficiently.

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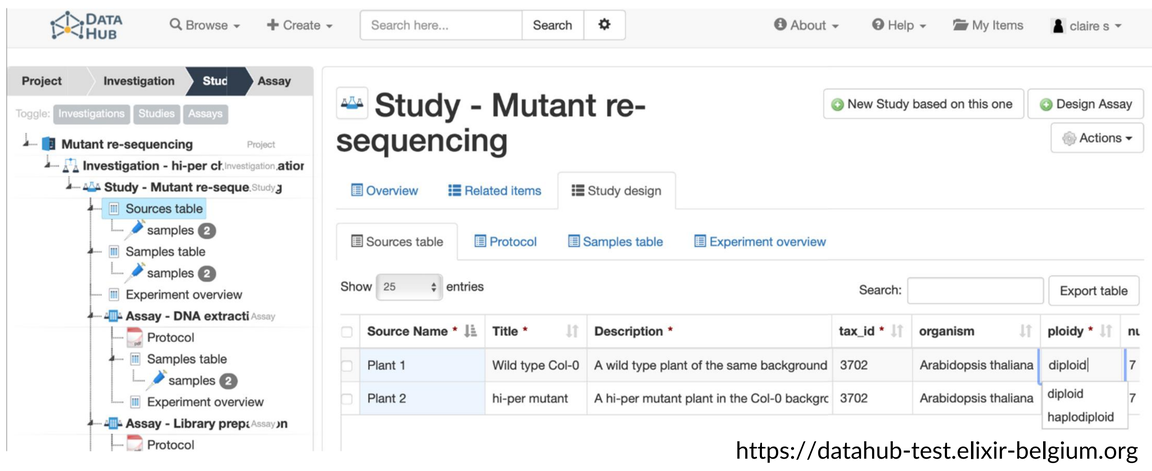
DataHub: FAIR (meta)data-by-design as a service

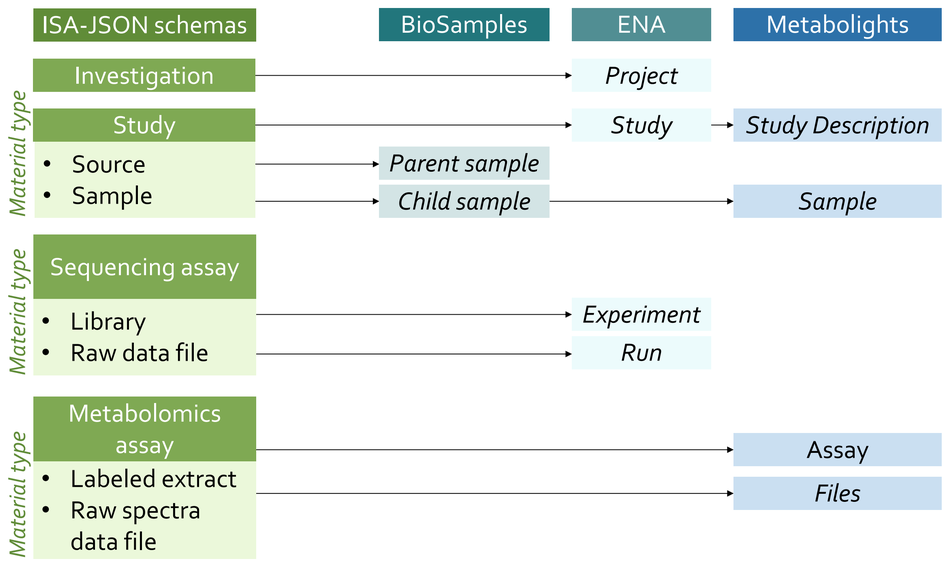


https://datahub-test.elixir-belgium.org

- For active (meta)data collection of experimental conditions and samples by researchers
  - Samples lineage
- Compliant with repositories' checklists and customisable
  - Controlled vocabularies
  - Ontologies from Ontology Lookup Service (OLS)
- Metadata export as ISA-JSON

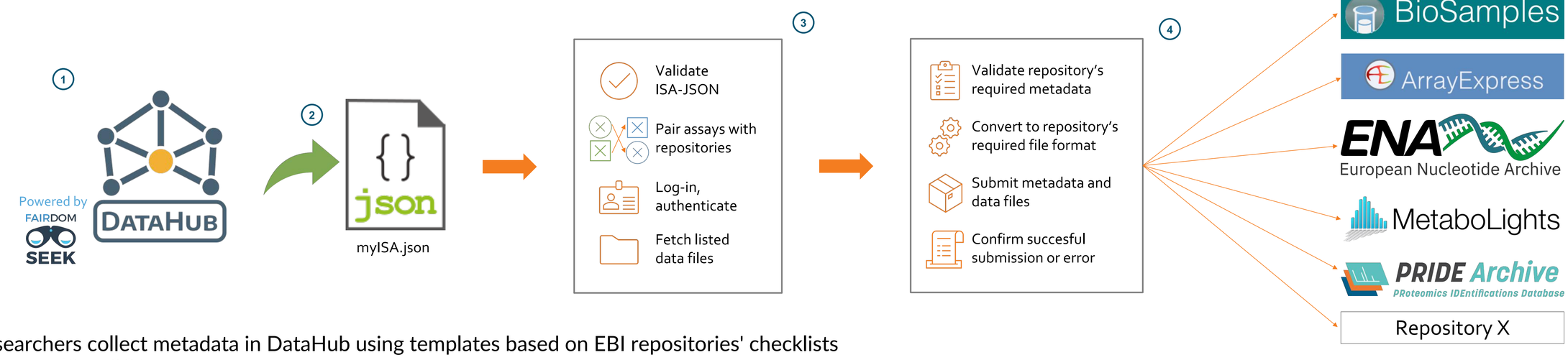
OLS

Data brokering tools: mapping ISA-JSON to EBI repositories metadata objects



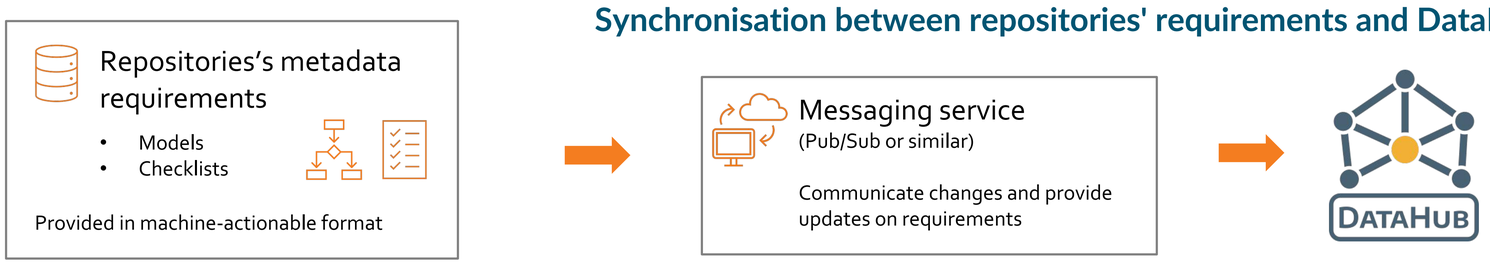
ISA-JSON file as starting point. This file contains multi-omics experiment metadata compliant with the requirements of the EBI repositories' metadata objects. This includes the Samples checklist of BioSamples and ENA (SRA) attributes, describing the library or the sequencing strategy utilized.

Multi-repository data submission using ISA-JSON



1. Researchers collect metadata in DataHub using templates based on EBI repositories' checklists
2. Export as ISA-JSON
3. An upload tool, independent of any specific repository, validates the ISA-JSON model and submits to the EBI multi-repository converter
4. The convertor maps and validates the ISA-JSON against the specific models of each repository and submits both data and metadata. Information about the submission is communicated back to the researcher

Synchronisation between repositories' requirements and DataHub templates



We aim to have an automated solution for converting repositories' requirements into DataHub templates. This system should rely on programmatic access to machine-actionable EBI repositories' checklists and metadata requirements.

Acknowledgement

Thanks to FAIRDOM and to all the participants in the project 'Streamlining data brokering from Research Data Management platforms to ELIXIR Repositories' during the Biohackathon 2022 for their contributions.
D'Anna, Flora, et al. "Streamlining Data Brokering from Research Data Management Platforms to ELIXIR Repositories." BioHackrXiv, 22 Feb. 2023. doi:10.37044/osf.io/mwk9f.

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DataHub development receives support from the Research Foundation – Flanders (FWO) under project No I002819N

These projects have received funding from the European Union's Horizon 2020 research and innovation programme

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